Ontology distance

$$Dist_O^{Onto}(t_1, t_2) = dist(t_1, lca_{t_1,t_2}) + dist(t_2, lca_{t_1,t_2})$$

$O \in \{HPO, BP, MF, CC\}$; *dist* = number of is_a edges to the lowest common ancestor; precomputed LCA matrices, not changed by weighting

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Optional term weights (shown for completeness; not used in the pipeline)

$$IC(t) = -\log \frac{n_t}{N}, \qquad \widehat{IC}(t) = \max\left(\varepsilon, \frac{IC(t)}{\max_{t'} IC(t')}\right)$$

n_t = DGS diseases annotated with t or any descendant of t (true-path rule); N = DGS diseases with ≥ 1 annotation on that axis; floor ε keeps all-hub diseases defined

$$w_{d,p} = \widehat{IC}(p)^\gamma \cdot e_{d,p} \qquad w_{d,g} = e_{d,g} \;\; [\cdot \widehat{IC}(g)^\gamma], \qquad \gamma \in \{1, 2\}$$

γ sharpens hub suppression: $\gamma = 1$ gives developmental delay 0.15 vs cherry red spot 0.66 (4x); $\gamma = 2$ gives 0.02 vs 0.44 (20x). Fever: 0.32 / 0.10

$$e_{d,p} = 0.5^{\,hops} \;\; [\cdot f_{d,p}] \qquad e_{d,g} = s(c_{d,g}) \;\; [\cdot q_{d,g}]$$

Symptom: $hops = 0$ for a direct HPOA annotation, ≥ 1 when inherited from a child disease; $f_{d,p}$ = HPOA frequency

Gene: $c_{d,g}$ = strongest source class of the edge (germline curated > OT aggregate > literature > GWAS solo, somatic solo); $s(c)$ = class prior, e.g. 1.0 / 1.0 / 1.0 / 0.5 / 0.5

$q_{d,g}$ = score percentile within its own source (scores are not comparable across sources); [ ] = optional factor

All $w = 1 \;\Rightarrow\; \sum_i w_i = m \;\Rightarrow$ the unweighted formulas exactly

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HPO distance

$$Dist_p(p_1, p_2) = \frac{Dist_{HPO}^{Onto}(p_1, p_2)}{d_{max}^{HPO}}$$

$d_{max}^{HPO} = 25$, diameter of HPO

$$Dist^{HPO}(d_1, d_2) = \frac{1}{2}\left(\frac{\sum_{i=1}^{m} w_{d_1,p_i} \min_{1 \leq j \leq n} Dist_p(p_i, p_j)}{\sum_{i=1}^{m} w_{d_1,p_i}} + \frac{\sum_{j=1}^{n} w_{d_2,p_j} \min_{1 \leq i \leq m} Dist_p(p_i, p_j)}{\sum_{j=1}^{n} w_{d_2,p_j}}\right)$$

m, n = number of d_1, d_2 symptoms; the weight belongs to the symptom being averaged, the best-match min is unweighted

With weights, $\frac{1}{m}\sum_i$ and $\frac{1}{n}\sum_j$ become weighted means; a shared generic symptom (e.g. developmental delay) contributes a 0 with small weight

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GO distance -- one per namespace (BP, MF, CC)

$$Dist_g^{BP}(g_1, g_2) = \frac{\min_{1 \leq i \leq n_{g_1}^{BP},\ 1 \leq j \leq m_{g_2}^{BP}} Dist_{BP}^{Onto}(g_1^i, g_2^j)}{d_{max}^{BP}}$$

$n_{g_1}^{BP}$, $m_{g_2}^{BP}$ = number of GO BP terms of g_1, g_2; $d_{max}^{BP} = 23$, diameter of GO biological process

$$Dist_g^{MF}(g_1, g_2) = \frac{\min_{1 \leq i \leq n_{g_1}^{MF},\ 1 \leq j \leq m_{g_2}^{MF}} Dist_{MF}^{Onto}(g_1^i, g_2^j)}{d_{max}^{MF}}$$

$n_{g_1}^{MF}$, $m_{g_2}^{MF}$ = number of GO MF terms of g_1, g_2; $d_{max}^{MF} = 18$, diameter of GO molecular function

$$Dist_g^{CC}(g_1, g_2) = \frac{\min_{1 \leq i \leq n_{g_1}^{CC},\ 1 \leq j \leq m_{g_2}^{CC}} Dist_{CC}^{Onto}(g_1^i, g_2^j)}{d_{max}^{CC}}$$

$n_{g_1}^{CC}$, $m_{g_2}^{CC}$ = number of GO CC terms of g_1, g_2; $d_{max}^{CC} = 18$, diameter of GO cellular component

Used in the pipeline: $n_{g_1}^X, m_{g_2}^X$ exclude hub-masked terms -- any GO term annotated to $> 2\%$ of its namespace's DGS genes

(e.g. "regulation of transcription by RNA pol II", on 900-1,100 DGS genes) is dropped from the gene-gene min before the formula runs; a masked-out gene keeps all its terms

This hard mask (hub fraction 0.02) is applied to BP, MF and CC alike

and is what produced the MechDist network used throughout

$$Dist_{g,\kappa}^X(g_1, g_2) = \min_{i,j}\left[(1-\kappa)\frac{Dist_X^{Onto}(g_1^i, g_2^j)}{d_{max}^X} + \kappa\left(1 - \widehat{IC}_X(lca_{ij})^\gamma\right)\right], \qquad \kappa = 0 \text{ gives the hard-masked formula above}$$

Alternative shown for completeness (not used): a soft, graded penalty in place of the hard mask: a gene pair whose best match runs through a generic-but-unmasked GO term

is downweighted rather than scored as identical. $\widehat{IC}_X$ is computed over DGS genes (true-path); lca_{ij} is read from the precomputed LCA matrix; for an exactly shared term $lca =$ the term itself

Hub GENES (e.g. TP53, on 152 diseases) would be handled at disease level by $\widehat{IC}(g)^\gamma$ in $w_{d,g}$ above (not used)

Hub GO TERMS (this box) are removed by the hard mask in the pipeline; κ is a finer alternative, not used

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$$Dist^X(d_1, d_2) = \frac{1}{2}\left(\frac{\sum_{i=1}^{m} w_{d_1,g_i} \min_{1 \leq j \leq n} Dist_g^X(g_i, g_j)}{\sum_{i=1}^{m} w_{d_1,g_i}} + \frac{\sum_{j=1}^{n} w_{d_2,g_j} \min_{1 \leq i \leq m} Dist_g^X(g_i, g_j)}{\sum_{j=1}^{n} w_{d_2,g_j}}\right), \quad X \in \{BP, MF, CC\}$$

m, n = number of d_1, d_2 genes annotated in namespace X; weights do not touch gene-gene $Dist_g^X$, so gene-gene distances are unchanged

$$Dist^{GO}(d_1, d_2) = \sum_{X \in present} \lambda_X Dist^X(d_1, d_2), \qquad \lambda_{BP} = 1.0,\ \lambda_{MF} = 0.25,\ \lambda_{CC} = 0.1$$

present = namespaces in which both diseases have annotated genes (a weighted sum, not renormalised); NaN if none

Optional (not used): gene weights $w_{d,g}$ in the average

Weights are normalised per disease, so $s(c)$ matters only for diseases that mix source classes: 359 of 9,609 DGS diseases with genes (curated + GWAS/somatic)

Black: formula; red: definitions and the pipeline as used; blue: optional weights and their effect. The GO hard hub-term mask is part of the pipeline; the term and gene weights and the soft penalty are shown for completeness and are not used.